

Unit-1

Lecturer Notes

Sub: Web Programming

Author: **M.B.S.Jadhav** M.Tech

Asst.Prof

E_Mail ID: bhavsinghm@gmail.com

Syllabus: HTML Common tags- List, Tables, images, forms, Frames; Cascading Style sheets;

Objectives:

- ✓ Understand common basic HTML Tags.
- ✓ Understand List Tags.
- ✓ Understand Form, Frames, CSS.

Introduction:

- ❖ HTML stands for Hypertext Markup Language.
- ❖ It is used to display the document in the web browsers.
- ❖ HTML pages can be developed to be simple text or to be complex multimedia program containing sound, moving images and java applets.
- ❖ HTML is considered to be the global publishing format for Internet.
- ❖ It is not a programming language. HTML was developed by Tim Berners-Lee.
- ❖ HTML standards are created by a group of interested organizations called W3C (world wide web consortium).
- ❖ In HTML formatting is specified by using tags. A tag is a format name surrounded by angle brackets.
- ❖ End tags which switch a format off also contain a forward slash. Points to be remembered for HTML tags:
 - ✓ Tags are delimited by angled brackets.
 - ✓ They are not case sensitive i.e., <head>, <HEAD> and <Head> is equivalent.
 - ✓ If a browser not understand a tag it will usually ignore it.
 - ✓ Some characters have to be replaced in the text by escape sequences.
 - ✓ White spaces, tabs and newlines are ignored by the browser.

Structure of an HTML document:

- ❖ All HTML documents follow the same basic structure.
- ❖ They have the root tag as <html>, which contains <head> tag and <body> tag.
- ❖ The head tag is used for control information by the browser and the body tag contains the actual user information that is to be displayed on the screen. The basic document is shown below.


```
<html>
<head>
<title> Basic HTML document </title>
</head>
<body>
<h1> Welcome to the world of Web Technologies</h1>
<p> A sample HTML program written by Amer </p>
</body>
</html>
```

Besides head and body tag, there are some other tags like title, which is a sub tag of head, that displays the information in the title bar of the browser. <h1> is used to display the line in its own format i.e., bold with some big font size. <p> is used to write the content in the form of paragraph. Comments in HTML documents start with <!-- and end with >. Each comment can contain as many lines of text as you like. If comment is having more lines, then each line must start and end with -- and must not contain -- within its body.

```
<!-- this is a comment line - -
-- which can have more lines - ->
```

Basic HTML tags

1. Body tag :

Body tag contain some attributes such as bgcolor, background etc. bgcolor is used for background color, which takes background color name or hexadecimal number and #FFFFFF and background attribute will take the path of the image which you can place as the background image in the browser.

<body bgcolor="#F2F3F4" background="c:\amer\imag1.gif">

2. Paragraph tag:

Most text is part of a paragraph of information. Each paragraph is aligned to the left, right or center of the page by using an attribute called as align. **<p align="left" | "right" | "center">**

3. Heading tag:

HTML is having six levels of heading that are commonly used. The largest heading tag is <h1> . The different levels of heading tag besides <h1> are <h2>, <h3>, <h4>, <h5> and <h6>. These heading tags also contain attribute called as align.

<h1 align="left" | "right" | "center"> <h2>

4. hr tag:

This tag places a horizontal line across the system. These lines are used to break the page. This tag also contains attribute i.e., width which draws the horizontal line with the screen size of the browser. This tag does not require an end tag.

<hr width="50%">.

5. base font:

This specify format for the basic text but not the headings.

<basefont size="10">

6. font tag:

This sets font size, color and relative values for a particular text.

7. bold tag:

This tag is used for implement bold effect on the text

** **

8. Italic tag:

This implements italic effects on the text.

<i>.....</i>

9. strong tag:

This tag is used to always emphasized the text

.....

10. tt tag:

This tag is used to give typewriting effect on the text

<tt>.....</tt>

11. sub and sup tag:

These tags are used for subscript and superscript effects on the text.

_{.....}

^{.....}

12. Break tag:

This tag is used to break the line and start from the next line.

**
**

13. < > "

These are character escape sequence which are required if you want to display characters that HTML uses as control sequences.

Example: **< can be represented as <.**

14. Anchor tag:

This tag is used to link two HTML pages, this is represented by <a>

** some text **

href is an attribute which is used for giving the path of a file which you want to link.

Example 1: HTML code to implement common tags.

mypage.html

```
<html>
<head> <!-- This page implements common html tags -->
<title> My Home page </title>
</head>
<body>
<h1 align="center"> GREEN FORT ENGINEERING COLLEGE</h1>
<h2 align="center"> Bandlaguda, Hyderabad</h2>
<basefont size=4>
<p> This college runs under the <tt>management</tt> of <font size=5> <b><i>&quot;
Syed Hashim Education Society&quot; & it is</i></b></font><br>
affiliated to <strong> JNTU</strong>
<hr size=5 width=80%>
<h3> <u>&lt;Some common tags&gt</u> </h3><br>
<a href="F:\feroz\wlab\prog1\list.html"> List </a><br>
<a href="F:\feroz\wlab\prog1\table.html"> Table </a><br>
</body>
</html>
```


Lists:

- ❖ One of the most effective ways of structuring a web site is to use lists. Lists provides straight forward index in the web site.
- ❖ HTML provides three types of list
 1. **Ordered(numbered) list**
 2. **Unordered List or(Bulleted List)**
 3. **Defination List**

****

✓ The ordered(numbered) and unordered(bulleted) lists are each made up of sets of list items. This tag is used to write list items

<ul type="disc" | "square" | "circle">

- ✓ This tag is used for basic unordered list which uses a bullet in front of each tag, every thing between the tag is encapsulated within tags.

<ol type="1" | "a" | "I" start="n">.....

- ✓ This tag is used for unordered list which uses a number in front of each list item or it uses any element which is mentioned in the type attribute of the tag, start attribute is used for indicating the starting number of the list.

<dl>..... </dl>

- ✓ This tag is used for the third category i.e., definition list, where numbers or bullet is not used in front of the list item, instead it uses definition for the items.

<dt>.....</dt>

- ✓ This is a sub tag of the <dl> tag called as definition term, which is used for marking the items whose definition is provided in the next data definition.

<dd></dd>

This is a sub tag of the <dd> tag, definition of the terms are enclosed within these tags.
The definition may include any text or block.

Example on List Tag

```
<html>
<head>
<title> list page </title>
</head>
<body>
```



```

<h3> Course details </h3><br>
<ul type="disc">
<li> Computer Science and Engineering </li>
<li> Information Technology </li>
<li> Electronics and Communication </li>
</ul><br>
<ol type="I" start=4>
<li> Mechanical Engineering</li>
<li> Electronics and Electrical Engineering</li>
</ol><br>
<h3> Subject Details </h3><br>
<dl>
<dt> Web Technologies</dt>
<dd> This subject is related to the technologies used in web applications</dd>
</dl>
<a href="d:\feroz\gfec\wtlab\as gm1.html">back</a>
</body>
</html>

```

Tables:

- ❖ Table is one of the most useful HTML constructs.
- ❖ Tables are found all over the web application. The main use of table is that they are used to structure the pieces of information and to structure the whole web page. Below are some of the tags used in table.

<table align="center" | "left" | "right" border="n" width="n%" cellpadding="n" cellspacing="n">.....</table>

- ❖ Every thing that we write between these two tags will be within a table. The attributes of the table will control in formatting of the table.
- ❖ Cell padding determines how much space there is between the contents of a cell and its border, cell spacing sets the amount of white space between cells.
- ❖ Width attribute sets the amount of screen that table will use.

<tr> </tr>

This is the sub tag of <table> tag, each row of the table has to be delimited by these tags.

<th>.....</th>

This is again a sub tag of the <tr> tag. This tag is used to show the table heading .

<td>.....</td>

This tag is used to give the content of the table.

Example 3: HTML code showing the use of table tag

```

<html>
<head>
<title> table page</title>

```



```

</head>
<body>
<table align="center" cellpadding="2" cellspacing="2" border="2">
<caption>Time for III year IT</caption>
<tr><th> I period </th>
<th> II peiord </th>
</tr>
<tr>
<td> wt </td>
<td> uml</td>
</tr>
</table>
</body>
</html>

```

Color and Image:

- ❖ Color can be used for background, elements and links. To change the color of links or of the page background hexadecimal values are placed in the <body> tag.

<body bgcolor = “#nnnnnn” text = “#nnnnnn” link= “#nnnnnn” vlink= “#nnnnnn” alink = “#nnnnnn”>

- ❖ The vlink attribute sets the color of links visited recently, alink the color of a currently active link. The six figure hexadecimal values must be enclosed in double quotes and preceded by a hash(#). Images are one of the aspect of web pages. Loading of images is a slow process, and if too many images are used, then download time becomes intolerable. Browsers display a limited range of image types.

<body background = “URL”>

This tag will set a background image present in the URL. Another tag that displays the image in the web page, which appears in the body of the text rather than on the whole page is given below

Example 4: HTML code that implements color and image

```

<html>
<head> <!-- This page implements color and image -->
<title> My Home page </title>
</head>
<body bgcolor="gray" text="magenta" vlink="yellow" alink="brown">
<img src= "d:\feroz\gfec\wtlab\asgm1.gif">
</body>
</html>

```


Example 5: HTML code that implements background image

```
<html>
<head> <!-- This page implements background image -->
<title> My Home page </title>
</head>
<body background="d:\feroz\gfec\wtlab\asgm1.gif">
</body>
</html>
```

Frames:

- ❖ Frames provide a pleasing interface which makes your web site easy to navigate.
- ❖ When we talk about frames actually we are referring to frameset, which is a special type of web page.
- ❖ The frameset contains a set of references to HTML files, each of which is displayed inside a separate frame.
- ❖ There are two tags related to frames i.e., **frameset** and **frame**

<frameset cols=" % , %" | rows=" % , %">.....</frameset>

<frame name="name" src="filename" scrolling =" yes" | "no" frameborder ="0"|"1">

Forms:

- ❖ Forms are the best way of adding interactivity of element in a web page.
- ❖ They are usually used to let the user to send information back to the server but can also be used to simplify navigation on complex web sites.
- ❖ The tags that use to implement forms are as follows.

<forms action="URL" method = "post" | "get">.....</form>

- ❖ When get is used, the data is included as part of the URL.
- ❖ The post method encodes the data within the body of the message.
- ❖ Post can be used to send large amount of data, and it is more secure than get. The tags used inside the form tag are:

<input type = "text" | "password" | "checkbox" | "radio" | "submit" name="string" value="string" size="n">

- ❖ In the above tag, the attribute type is used to implement text, password, checkbox, radio and submit button.

Text:

- ✓ It is used to input the characters of the size n and if the value is given than it is used as a default value.
- ✓ It uses single line of text. Each component can be given a separate name using the name attribute.

Password:

- ✓ It works exactly as text, but the content is not displayed to the screen, instead an * is used.

Radio:

- ✓ This creates a radio button. They are always grouped together with a same name but different values.

Checkbox:

- ✓ It provides a simple checkbox, where all the values can be selected unlike radio button.

Submit:

- ✓ This creates a button which displays the value attribute as its text. It is used to send the data to the server.

<select name="string">.....</select>

- ✓ This tag helps to have a list of item from which a user can choose. The name of the particular select tag and the name of the chosen option are returned.

<option value="string" selected>.....</option>

- ✓ The select statement will have several options from which the user can choose. The values will be displayed as the user moves through the list and the chosen one returned to the server.

<textarea name="string" rows="n" cols="n">.....</textarea>

- ✓ This creates a free format of plain text into which the user can enter anything they like. The area will be sized at rows by cols but supports automatic scrolling.

Example 6: HTML code that implements forms

```
<html>
<head>
<title>form</title>
```



```

</head>
<body>
<p align="left">Name:<input type="text" maxlength=30 size=15>
<p align="left">Password:<input type="password" maxlength=10 size=15>
<p align="left">Qualification: <br><input type="checkbox" name="q" value="be">B.E
<input type="checkbox" name="q" value="me">M.E
<p align="left">Gender:<br> <input type="radio" name="g" value="m">Male
<input type="radio" name="g" value="f">Female
<p align="left">course:<select name="course" size=1>
<option value=cse selected>CSE
<option value=it>CSIT
</select>
<p align="left">Address:<br><textarea name="addr" rows=4 cols=5
scrolling=yes></textarea>
<p align="center"><input type="submit" name="s" value="Accept">
<p align="center"><input type="reset" name="c" value="Ignore">
</body>
</html>

```

Example 7: HTML code that implements frames

```

<html>
<head>
<title> My page </title>
</head>
<frameset rows="25%,50%">
<frame name="a" src="f:\feroz\wlab\asgm1.html">
<frameset cols="25%,50%">
<frame name="b" src="f:\feroz\wlab\index.html">
<frame name="abc" src="f:\feroz\wlab\welcome.html">
</frameset>
</frameset>
</html>

```

Cascading Stylesheets:

- ❖ One of the most important aspects of HTML is the capability to separate presentation and content.
- ❖ A style is simply a set of formatting instructions that can be applied to a piece of text. There are three mechanisms by which we can apply styles to our HTML documents. · Style can be defined within the basic HTML tag.

💡 Style can be defined in the <head> tag

- ✓ Styles can be defined in external files called stylesheets which can then be used in any document by including the stylesheet via a URL.
- ✓ A style has two parts: a selector and a set of declarations. The selector is used to create a link between the rule and the HTML tag.

The declaration has two parts:

- ✓ A property and a value. Declarations must be separated using colons and terminated using semicolons.

Selector{property: value; property: value}

Properties and values in styles:

Fonts:

Font-family: <family name.
 Font-style: normal | italic | oblique
 Font-weight: normal | bold | bolder | lighter
 Font-size: small | medium | large | smaller | larger

Backgrounds and Colors

Color: value
 Background-color: value
 Background-image: URL

Text

Text-decoration: none | underline | overline
 Text-transformation: none | uppercase | lowercase
 Text-align: left | right | center | justify
 Text-indentation : length | percentage

Example 7: HTML code representing cascading style sheet


```
<html>
<head>
<title>My Web Page</title>
<style type="text/css">
h1 {font-family:mssanserif;font-size:30;font-style:italic;fontweight:
bold;color:red;background-color:blue;border:thin groove}
.m{border-width:thick;border-color:red;border-style:dashed}
.mid {font-family:BankGothicLtBT;text-decoration:link;texttransformation:uppercase;
text-indentation:60%}
</style>
</head>
<body class="m">
<h1> Green Fort Engineering College</h1>
<p class="mid">Jawaharlal Technological University Hyderabad</p>
</div>
</body>
</html>
```